



Subquery in SQL – Detailed Examples

◆ 1. Subquery with **SELECT** Statement

Use case: Retrieve data based on a condition derived from another query.

```
-- Get names of employees who earn more than the average salary
SELECT name
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

◆ 2. Subquery with **INSERT** Statement

Use case: Insert data retrieved from another query.

```
-- Insert all top-performing students into the honors table
INSERT INTO honors_students (student_id, name)
SELECT id, name
FROM students
WHERE gpa > (SELECT AVG(gpa) FROM students);
```

◆ 3. Subquery with **UPDATE** Statement

Use case: Update values in a table using data from another table/query.

```
-- Give a 10% salary hike to employees who work in the 'R&D' department
UPDATE employees
SET salary = salary * 1.10
WHERE department_id = (
    SELECT id FROM departments WHERE name = 'R&D'
);
```

◆ 4. Subquery with **DELETE** Statement

Use case: Delete records that match a condition fetched from another query.

```
-- Delete orders placed by inactive customers
DELETE FROM orders
WHERE customer_id IN (
    SELECT id FROM customers WHERE status = 'inactive'
);
```

◆ 5. Subquery inside **WHERE** Clause

Use case: Use subquery to filter rows with a condition.

```
-- Get names of students enrolled in course 'Database Systems'
SELECT name
FROM students
WHERE id IN (
    SELECT student_id
    FROM enrollments
    WHERE course_id = (
        SELECT id FROM courses WHERE name = 'Database Systems'
    )
);
```

◆ 6. Subquery with Operators

◆ = Operator

```
SELECT name FROM employees
WHERE department_id = (SELECT id FROM departments WHERE name = 'HR');
```

◆ < or > Operator

```
SELECT name FROM products
WHERE price < (SELECT AVG(price) FROM products);
```

◆ IN Operator

```
SELECT name FROM students
WHERE id IN (
    SELECT student_id FROM honors_list
);
```

◆ BETWEEN Operator

```
SELECT name FROM students
WHERE gpa BETWEEN (
    SELECT MIN(gpa) FROM students WHERE grade = 'B'
) AND (
    SELECT MAX(gpa) FROM students WHERE grade = 'B'
);
```
